DevOps

Enterprise Application Development

IFS

Agenda

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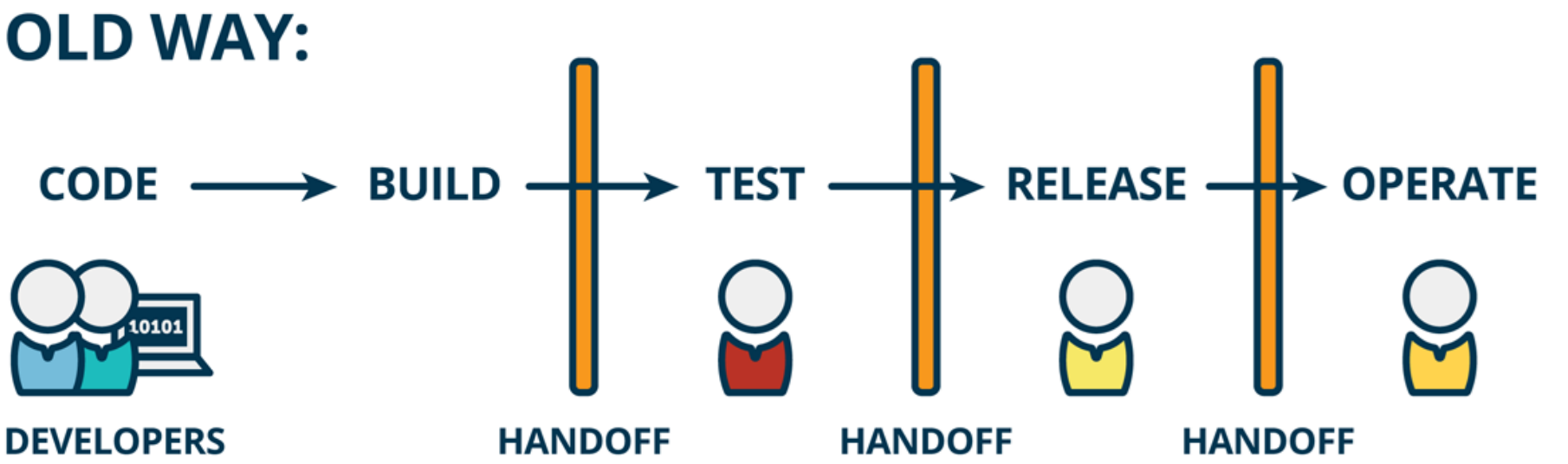
Traditional Software Development Practice



Software Engineers	Software Engineers	QA Engineers	System Engineers

Traditional Software Development Practice

Issues with Traditional method

- Deliveries are usually very large due to big chunks of code
- Usually, lot of manual process to be involved and slow progress
- Takes long time to deliver a new features and bug fixes
- Developer environments, build environments and test and production environments are not identical
- Due to mismatches of environments unexpected errors can occur in each hand off phase
- Developers may not be aware of build lifecycle and testing aspects

What is DevOps

DevOps is a set of practices that combines software development and IT operations

Definitions

“Union of people, processes, and products to enable continuous delivery of value to end users” - Microsoft

“DevOps is the combination of cultural philosophies, practices, and tools that increases an organization’s ability to deliver applications and services at high velocity” – Amazon

Quality assurance and security teams may also become more tightly integrated with development and operations in application lifecycle

DevOps + Security - > DevSecOps

DevOps

DevOps Practice

DevOps

DevOps Practice

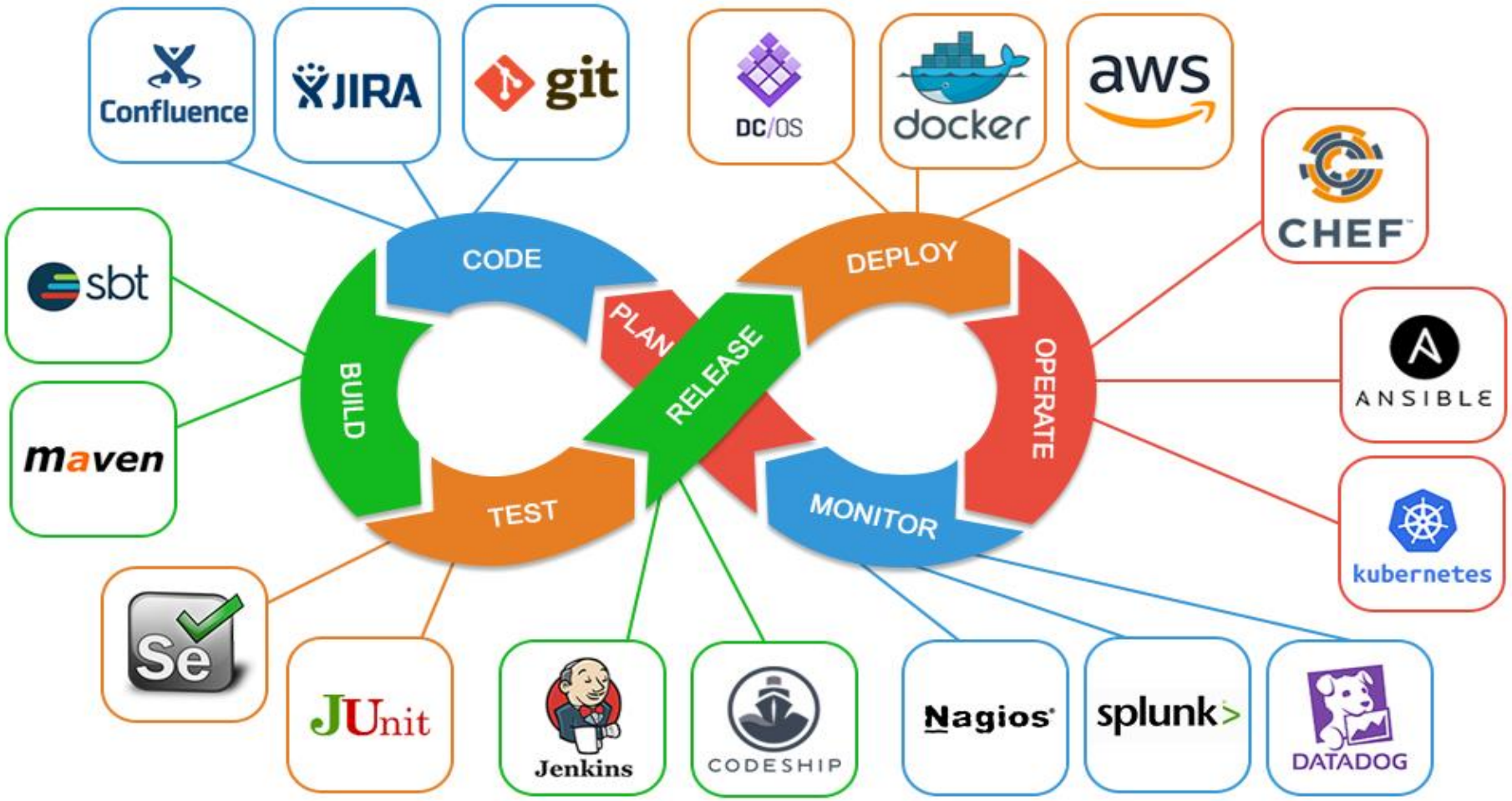
- Deliveries are usually very small and frequent
- Almost all the steps are automated using various tools
- Takes shorter time to deliver a functionality and fixes
- Developer environments, build environments and test and production environments are identical
- Therefore, delivery process hardly fails due to environment specific issues
- Developers are aware and capable of build lifecycle and testing aspects

DevOps

Adopting to the DevOps model

- DevOps is about removing the barriers between development and operations
- With DevOps multiple teams can work together to optimize productivity of developers and reliability of operations
- Transitioning to DevOps requires a change in culture and mindset
- In addition to that it needs several tools to automate most of the development and operational phases which were traditionally done as a repetitive manual processes

DevOps phases and tools



DevOps phases and tools

Code – Git

Build - Ant, Maven, Gradle

Test - JUnit, Selenium, Cucumber, JMeter

Release - Jenkins, Bamboo, Hudson, GitLab

Deploy - Docker, Azure, AWS, Terraform, Chef, Ansible

Operate - Puppet, Chef, Ansible

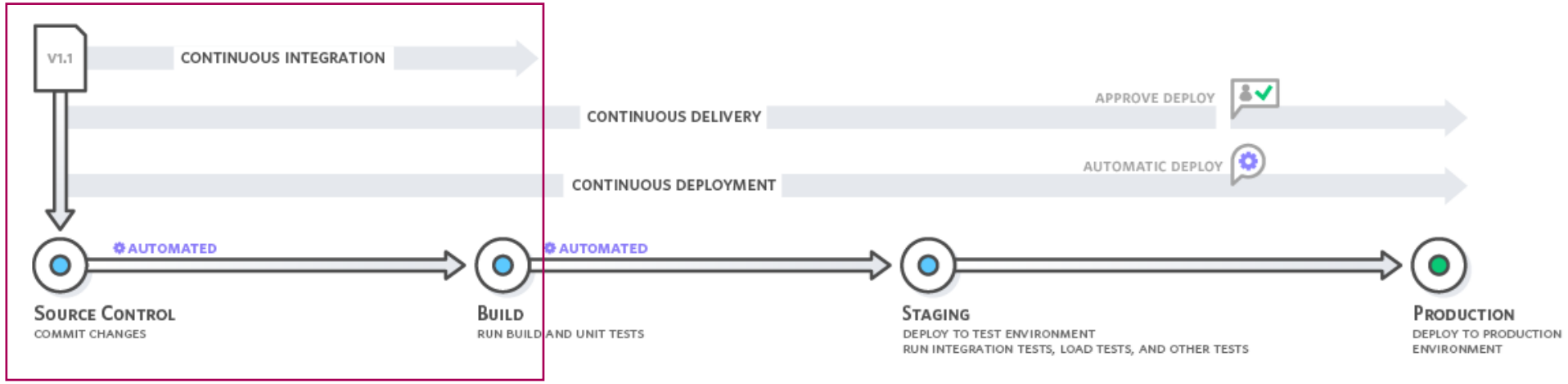
Monitor - Nagios, Stack, Splunk, Grafana

Phases of DevOps Lifecycle

- Continuous Integration
- Continuous Delivery
- Continuous Deployment
- Monitoring and Logging
- Automated Testing
- Infrastructure as code

Phases of DevOps Lifecycle

Continuous Integration



- Developers regularly merge code changes into a central repository and every such revision committed triggers an automated build and test.
- Number of tools are available which can pull the checked in code to prepare a build

Phases of DevOps Lifecycle

Continuous Integration

- Increase productivity of developers
    - Developers are free from repetitive manual building and testing tasks.
    - They can invest their time/code to write new functionality when the automated tests discover bugs.

- Quick finding and resolving bugs
    - Frequent testing helps to discover bugs earlier before they grow into larger problems.

- Ability to deliver updates faster
    - Frequent bug free releases contributes to shorten the time taken to deliver the product.

Phases of DevOps Lifecycle

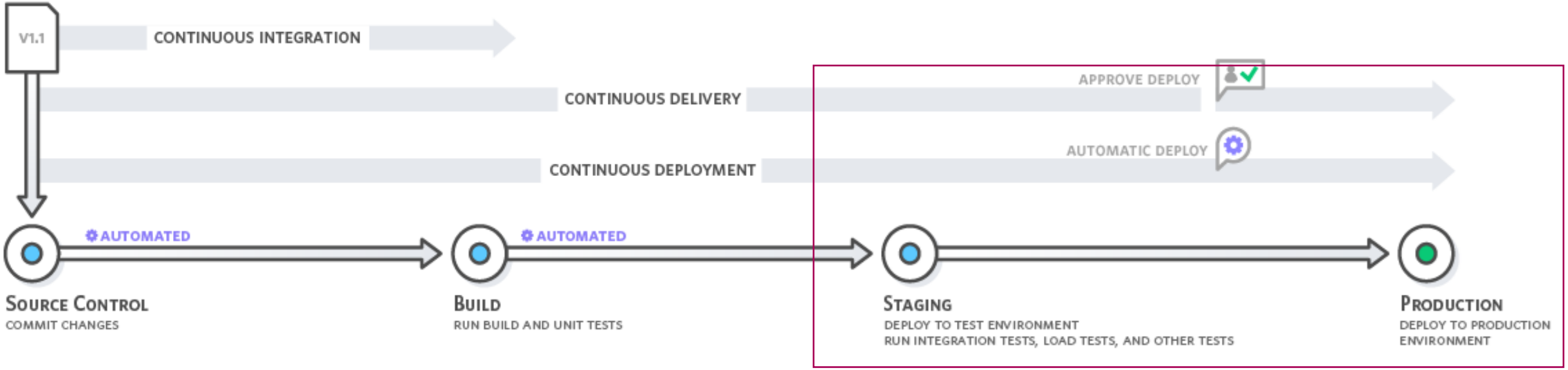
Continuous Delivery

- The continuous delivery process deploy the deliveries to a series of test environments where last environment being installed should be a mirror of a production environment.
- Ability to perform the User Acceptance Testing
- Builds up the team's confidence to apply it in the production environment

- Benefits
  - Faster delivery of new features and bug fixes
  - Faster software release process
  - Increase the productivity of the developers
  - Faster Mean Time To Resolution (MTTR)
  - Reduce costs

Phases of DevOps Lifecycle

Continuous Deployment



Perform the same tasks of Continuous Delivery and go further by automatically deploying the release into the actual production environment.

Here the delivering to production environment happens automatically without explicit approval.

Phases of DevOps Lifecycle

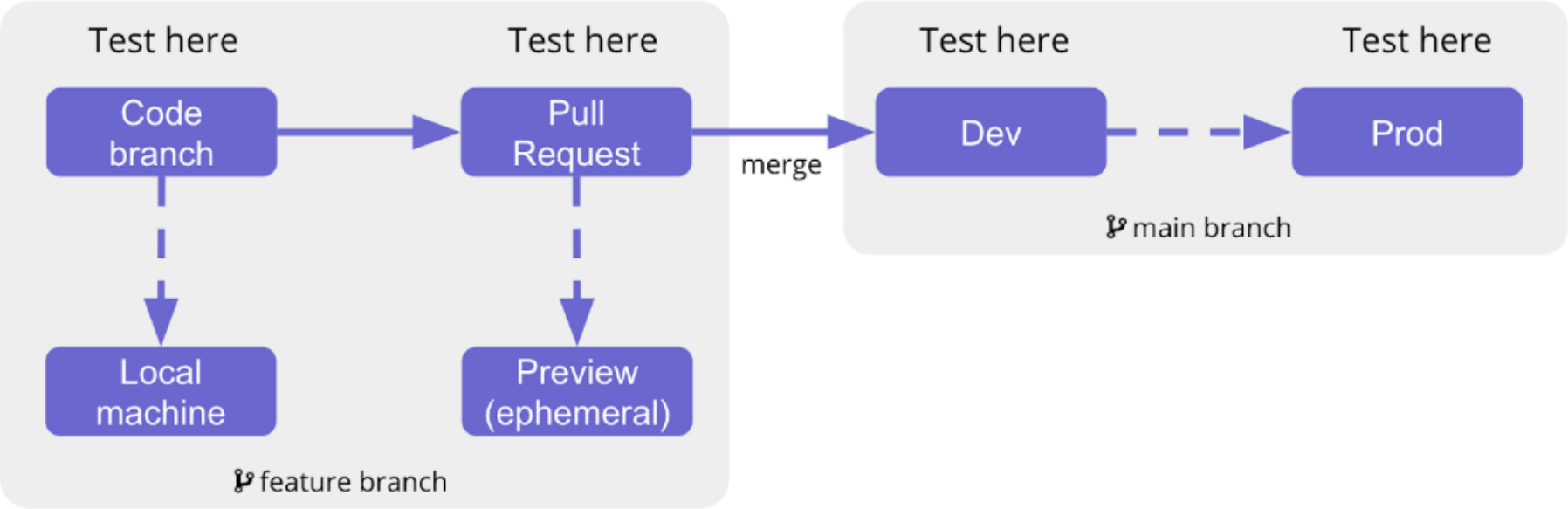
Monitoring and Logging

- DevOps monitoring entails overseeing the entire development process from planning, development, integration and testing, deployment, and operations
- It involves a complete and real-time view of the status of applications, services, and infrastructure in the production environment
- Features such as real-time streaming, historical replay, and visualizations are critical components of application and service monitoring

- Shift-left testing
- Alert and incident management

Phases of DevOps Lifecycle

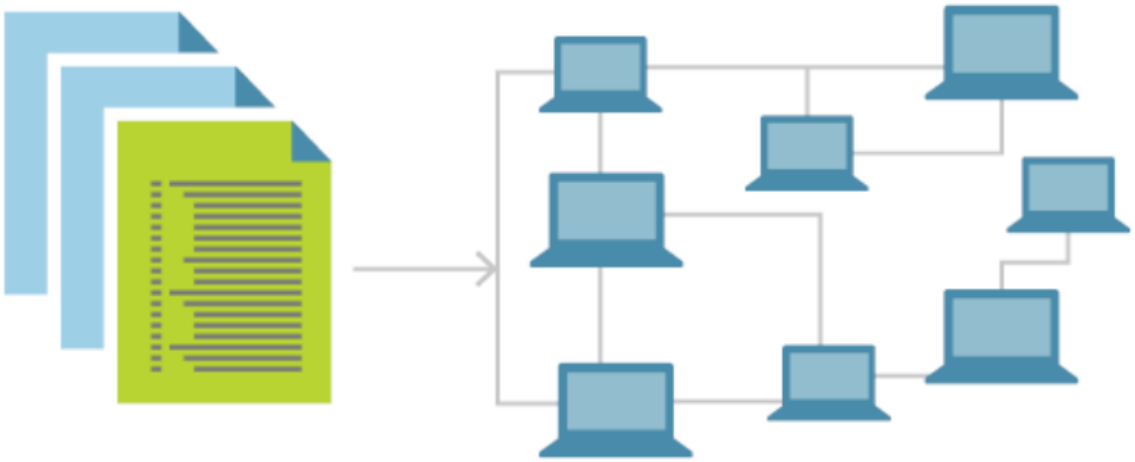
Automated Testing



- A DevOps best practice is to run automated tests as early and as often as possible within the CI/CD pipeline
- Run automated UI tests in production to proactively monitor for user experience issues
- Get the ability to early detection of issues

Phases of DevOps Lifecycle

Infrastructure as code



Infrastructure as Code (IaC) is the managing and provisioning of infrastructure through code instead of through manual processes

- Keep infrastructure definitions as source code files
- Version controlled
- Easily create and delete infrastructure resources
- No manual work

Thank you!

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